

Lecture Scribe Generation Request

L20

 PDF

Example Le...Scribe (1)

 PDF

Using the scribe structure pdf provided in the attached "Example Lecture Scribe" PDF, generate a lecture scribe for the provided Lecture 20 (mention lecture number) materials. The scribe must have all the topics included and its content based on example pdf. This scribe is intended to serve as a reference material for a closed-notes exam preparation. You must prioritize correctness and maintain strict faithfulness to the provided context, including definitions, notation, theorem statements, and step-by-step proofs (if any), worked examples exactly as they appear in the slides. Identify every major topic in the lecture, For EACH identified topic, you must independently repeat the given structure in example scribe format. * Do not include any creative explanations, external intuition, or material not traceable to the attached files*. Organize the output to follow the logical flow and scope of the lecture to ensure it is suitable for reliable revision.



Lecture Scribe: CSE 400 Lecture 20

Topic: Linear Transformations and Expectations of Multiple RVs **Scribe:**
Fagun Rathod - AU2440111

1. Random Vector Representation

2. Definitions and Notation

- **Random Vector:** For random variables $X_1, X_2, \dots, X_N$, they are represented as a vector $\underline{X}$.

$$\underline{X} = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_N \end{bmatrix}.$$

- **Purpose:** Provides a compact representation of multiple random variables.
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2. Statistical Representation: Mean Vector

2. Definitions and Notation

- **Mean Vector Definition:** The mean vector of $\underline{X}$ is $\underline{\mu}_X = E[\underline{X}]$.

$$\underline{\mu}_X = \begin{bmatrix} E[X_1] \\ E[X_2] \\ \vdots \\ E[X_N] \end{bmatrix}.$$

3. Assumptions / Conditions

- Each element in the mean vector corresponds to the mean of one individual random variable.
 - It represents the average value of the random vector.
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3. Statistical Representation: Correlation Matrix

2. Definitions and Notation

- **Correlation Matrix (R_{XX}):** Defined as $R_{XX} = E[\underline{X}\underline{X}^T]$.
- **Element Definition:** $R_{XX}(i, j) = E[X_i X_j]$ for $i, j \leq N$.
- **Matrix Structure:** $R_{XX} =$

$$\begin{bmatrix} E[X_1 X_1] & E[X_1 X_2] & \dots & E[X_1 X_N] \\ E[X_2 X_1] & \dots & E[X_2 X_N] & \\ \vdots & \vdots & \vdots & \\ E[X_N X_1] & E[X_N X_2] & \dots & E[X_N X_N] \end{bmatrix}_{N \times N}.$$

3. Assumptions / Conditions

- R_{XX} is a symmetric matrix.

4. Statistical Representation: Covariance Matrix

2. Definitions and Notation

- Covariance Matrix (C_{XX}):** For a random vector $\underline{X}$, $C_{XX} = E[(\underline{X} - \underline{\mu}_X)(\underline{X} - \underline{\mu}_X)^T]$.
- Practical Structure:** $C_{XX} =$

$Var(X_1)$ $Cov(X_1, X_2)$ $\dots$

$[Cov(X_2, X_1)$ $Var(X_2)$ $\dots]$

$\vdots$ $\vdots$ $\dots$

3. Assumptions / Conditions

- Symmetry:** $Cov(X_i, X_j) = Cov(X_j, X_i)$, making the matrix a "mirror" of itself.

5. Linear Transformations of Random Vectors

2. Definitions and Notation

- Linear Model:** $\underline{Y} = \underline{A}\underline{X} + \underline{b}$.
 - $\underline{A}$: Transformation matrix (scales and rotates data).
 - $\underline{b}$: Constant vector (shifts/translates the center).
 - $\underline{X}$: Input random vector.

4. Main Results / Theorems

Property	Transformed Result
Mean	$\mu_v = A\mu_x + b$

Property	Transformed Result
Covariance	$C_{\mathbf{v}\mathbf{v}} = \mathbf{A}C_{\mathbf{x}\mathbf{x}}\mathbf{A}^T$
Correlation	$R_{\mathbf{v}\mathbf{v}} = \mathbf{A}R_{\mathbf{x}\mathbf{x}}\mathbf{A}^T + \mathbf{A}\mu_{\mathbf{x}}\mathbf{b}^T + \mathbf{b}\mu_{\mathbf{x}}^T\mathbf{A}^T + \mathbf{b}\mathbf{b}^T$
	

5. Proofs / Derivations

Mean Transformation Derivation

- **Step 1:** $\mu_Y = E[\underline{Y}]$.
- **Step 2:** Substitute $\underline{Y} = \underline{A}\underline{X} + \underline{b}$ into the expectation: $\mu_Y = E[\underline{A}\underline{X} + \underline{b}]$.
- **Step 3:** Apply linearity: $E[\underline{A}\underline{X}] + E[\underline{b}]$.
- **Step 4:** Result: $\mu_Y = \underline{A}\mu_X + \underline{b}$.

Correlation Matrix Transformation Derivation

- **Step 1:** $R_{YY} = E[\underline{Y}\underline{Y}^T] = E[(\underline{A}\underline{X} + \underline{b})(\underline{A}\underline{X} + \underline{b})^T]$.
- **Step 2:** Expansion: $R_{YY} = E[\underline{A}\underline{X}\underline{X}^T\mathbf{A}^T + \underline{A}\underline{X}\mathbf{b}^T + \mathbf{b}\underline{X}^T\mathbf{A}^T + \mathbf{b}\mathbf{b}^T]$.
- **Step 3:** Applying Expectation: $R_{YY} = \underline{A}E[\underline{X}\underline{X}^T]\mathbf{A}^T + \underline{A}E[\underline{X}]\mathbf{b}^T + \mathbf{b}E[\underline{X}^T]\mathbf{A}^T + \mathbf{b}\mathbf{b}^T$.
- **Step 4:** Result: $R_{YY} = \underline{A}R_{XX}\mathbf{A}^T + \underline{A}\mu_X\mathbf{b}^T + \mathbf{b}\mu_X^T\mathbf{A}^T + \mathbf{b}\mathbf{b}^T$.

Covariance Matrix Transformation Derivation

- **Step 1:** Observe $\underline{Y} - \mu_Y = (\underline{A}\underline{X} + \underline{b}) - (\underline{A}\mu_X + \underline{b}) = \underline{A}(\underline{X} - \mu_X)$.
- **Step 2:** $C_{YY} = E[(\underline{Y} - \mu_Y)(\underline{Y} - \mu_Y)^T]$.

- **Step 3:** Substitute Step 1: $C_{YY} = E[\underline{A}(\underline{X} - \mu_X)(\underline{X} - \mu_X)^T \underline{A}^T]$.
- **Step 4:** Result: $C_{YY} = \underline{A}C_{XX}\underline{A}^T$.

6. Worked Examples

Example 1: Numerical Linear Transformation

Given: $\underline{Y} = \underline{A}\underline{X}$ where:

$$\underline{\mu}_x = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}, \underline{A} = \begin{bmatrix} 1 & 0 & -1 \\ 0 & -1 & 1 \\ -1 & 1 & 0 \end{bmatrix}, \underline{R}_{XX} = \begin{bmatrix} 13 & 2 & 3 \\ 2 & 10 & 3 \\ 3 & 3 & 10 \end{bmatrix}.$$

- **(a) Mean vector of Y:**

$$\underline{\mu}_Y = \underline{A}\underline{\mu}_X = \begin{bmatrix} 1 & 0 & -1 \\ 0 & -1 & 1 \\ -1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix}.$$

- **(b) Correlation matrix of Y:**

$$\underline{R}_{YY} = \underline{A}\underline{R}_{XX}\underline{A}^T.$$

$$\text{Step 1: } \underline{A}\underline{R}_{XX} = \begin{bmatrix} 10 & -1 & -7 \\ 1 & -7 & 7 \\ -11 & 8 & 0 \end{bmatrix}.$$

$$\text{Step 2: Multiply by } \underline{A}^T \rightarrow \underline{R}_{YY} = \begin{bmatrix} 17 & -6 & -11 \\ -6 & 14 & -8 \\ -11 & -8 & 19 \end{bmatrix}.$$

- **(c) Covariance matrix of Y:**

$$\text{Use } \underline{C}_{YY} = \underline{R}_{YY} - \underline{\mu}_Y \underline{\mu}_Y^T.$$

$$\text{Outer product: } \underline{\mu}_Y \underline{\mu}_Y^T = \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix} \begin{bmatrix} 1 & 2 & -3 \end{bmatrix} = \begin{bmatrix} 1 & 2 & -3 \\ 2 & 4 & -6 \\ -3 & -6 & 9 \end{bmatrix}.$$

$$\text{Subtraction: } \underline{C}_{YY} = \begin{bmatrix} 17 & -6 & -11 \\ -6 & 14 & -8 \\ -11 & -8 & 19 \end{bmatrix} - \begin{bmatrix} 1 & 2 & -3 \\ 2 & 4 & -6 \\ -3 & -6 & 9 \end{bmatrix} = \begin{bmatrix} 16 & -8 & -8 \\ -8 & 10 & -2 \\ -8 & -2 & 10 \end{bmatrix}.$$

6. Multivariate Gaussian Distribution

2. Definitions and Notation

- **Univariate (1D):** $f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$.
- **Joint Gaussian (2D):** $f_{XY}(x, y) = \frac{\exp\left\{-\frac{1}{2(1-\rho_{XY}^2)}\left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 - 2\rho_{XY}\frac{(x-\mu_X)(y-\mu_Y)}{\sigma_X\sigma_Y} + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2\right]\right\}}{\sqrt{2\pi\sigma_X\sigma_Y} \sqrt{1-\rho_{XY}^2}}$.
- **N-Dimensional Vector PDF:** $f_{\underline{X}}(\underline{x}) = \frac{1}{(2\pi)^N \det(C_{XX})} \exp\left(-\frac{1}{2}(\underline{x} - \underline{\mu}_X)^T C_{XX}^{-1} (\underline{x} - \underline{\mu}_X)\right)$.

3. Assumptions / Conditions

- $\underline{\mu}_X$: Mean vector ($N \times 1$).
- C_{XX} : Covariance matrix ($N \times N$).

4. Main Results / Theorems

- **Uncorrelated Gaussian RVs:** If a vector of N jointly Gaussian random variables is mutually uncorrelated ($Cov(X_i, X_j) = 0 \forall i \neq j$), the covariance matrix C_{XX} is diagonal.
- **Factorization:** The joint PDF factorizes into the product of individual Gaussian PDFs: $f_{\underline{X}}(\underline{x}) = \prod_{n=1}^N \frac{1}{\sqrt{2\pi}\sigma_n} \exp\left(-\frac{(x_n - \mu_n)^2}{2\sigma_n^2}\right)$.

6. Worked Examples

Example 2: Proving the 2D Joint Gaussian PDF

Given: $\underline{\mu}_X = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}$, $C_{XX} = \begin{bmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{bmatrix}$.

- **Step 1: Determinant:** $\det(C_{XX}) = \sigma_1^2\sigma_2^2(1 - \rho^2)$.
- **Step 2: Inverse Matrix:** $C_{XX}^{-1} = \frac{1}{\sigma_1^2\sigma_2^2(1-\rho^2)} \begin{bmatrix} \sigma_2^2 & -\rho\sigma_1\sigma_2 \\ -\rho\sigma_1\sigma_2 & \sigma_1^2 \end{bmatrix}$.
- **Step 3: Quadratic Form:** $Q(\underline{x}) = (\underline{x} - \underline{\mu}_X)^T C_{XX}^{-1} (\underline{x} - \underline{\mu}_X)$.
- **Step 4: Result:** $Q(\underline{x}) = \frac{1}{\sigma_1^2\sigma_2^2(1-\rho^2)} \left[\sigma_2^2 \left(\frac{x_1 - \mu_1}{\sigma_1}\right)^2 - 2\rho \left(\frac{x_1 - \mu_1}{\sigma_1}\right) \left(\frac{x_2 - \mu_2}{\sigma_2}\right) + \sigma_1^2 \left(\frac{x_2 - \mu_2}{\sigma_2}\right)^2 \right]$.